

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building full-stack features (Next.js + Firebase/GCP) on a provider-facing **SMART on FHIR** clinical decision support app, taking them from design through implementation with autonomy on an early-stage team.
- Extending a serverless **Firebase Cloud Functions** backend that assembles medication alternatives from **RxNorm** normalization and clinical-guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls results, keeping AI output verifiable rather than a black box.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's text-analysis platform (**Python, FastAPI, React**) end to end, hired off a working prototype built before the interview and given ownership of architecture and all technical decisions.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system; grew the backend to **40 hermetic tests** running in CI without ML dependencies.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Owned backend security and reliability for the multi-tenant stack, implementing **AWS KMS** encryption for sensitive data, hardening authentication and rate limiting, and remediating **20+ VAPT findings** with regression-tested fixes.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput **60%**.

Projects

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native AI research assistant end to end on **Cloudflare Workers**: an agent fetches web sources and streams multi-step research digests over WebSocket, with durable workflows (**WorkflowEntrypoint + Durable Objects**) that survive process restarts and tool failures and **D1** for state, no external database.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, deterministic extraction with LLM fallback, and **cell-level provenance**; hardened with **150+ automated tests** and provider routing/fallback across LLM backends.

Cache- and SIMD-Aware Matrix Multiplication (GitHub) C++, AVX/SIMD, OpenMP, Linux perf

- Optimized C++ matrix multiplication from a naive baseline through cache-aware tiling, manual **AVX/SIMD** vectorization via `__m256` intrinsics, register-aware micro-kernels, and **OpenMP** multithreading, profiling L1/L3/TLB misses with perf.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: TypeScript, JavaScript, Python, C++, Go, Java, SQL, Bash

Backend & Web: Node.js, Express, FastAPI, React, Next.js, REST APIs, multi-tenant architecture

Edge & Systems: Cloudflare Workers, Workers AI, Durable Objects, D1, WebSockets, Linux perf, AVX/SIMD

AI & LLM: LLM agents, RAG, provider routing/fallback, cell-level provenance, PyTorch, sentence-transformers

Cloud & DevOps: AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.